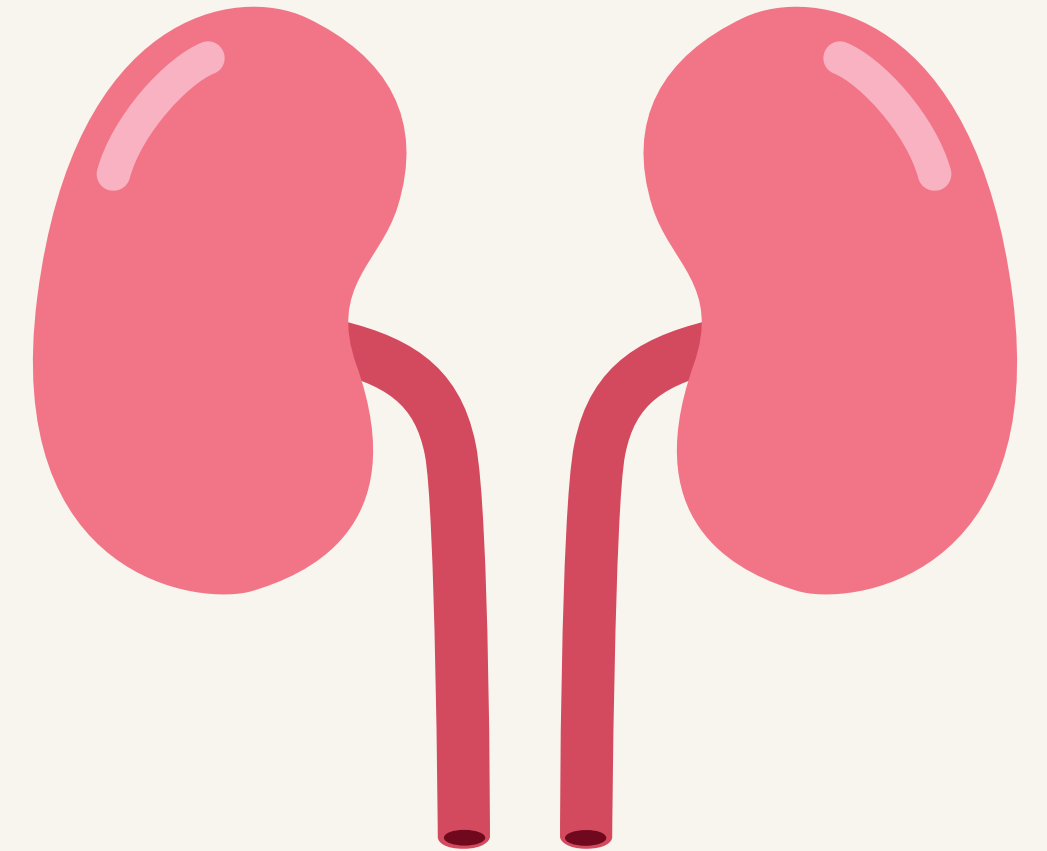


Chronic Kidney Disease (CKD)



How a patient's journey differs with
Social Vulnerability Index

Team Y-Star: Mariana Castellanos, Mikey Chen, Dan Le, Siddharth Chittapuram

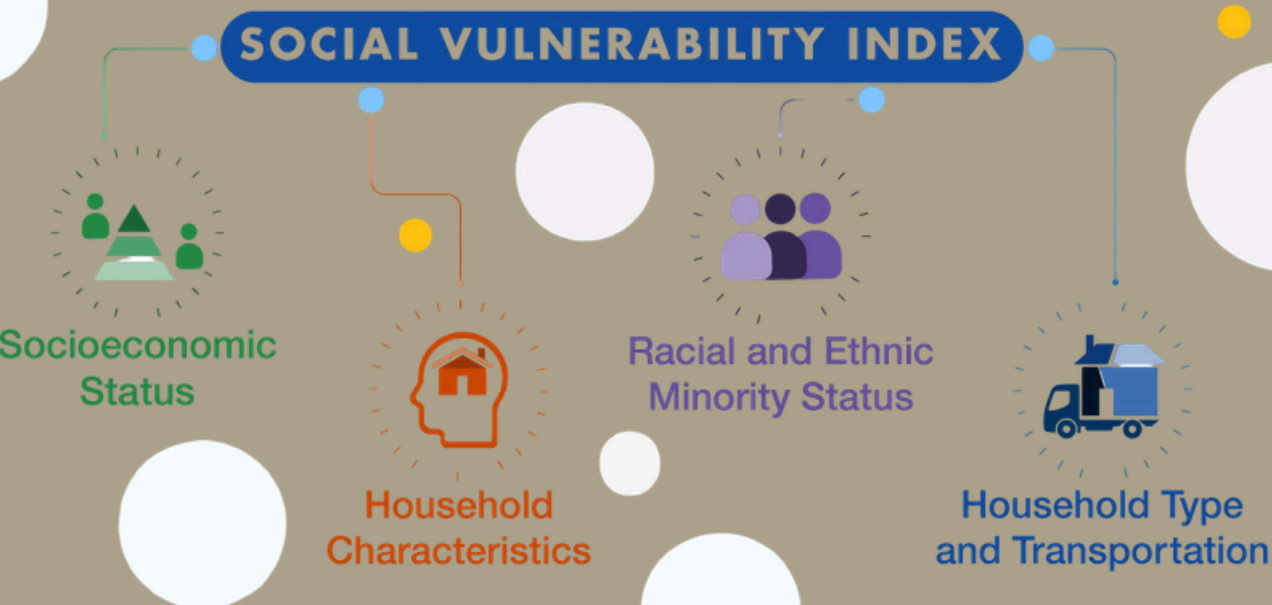
MAY 3RD 2026

Why it matters

- Kansas ranks **14th in the U.S.** for chronic kidney disease among adults
- Kidney Disease is Kansas's **#9 leading cause of death.**
- CKD patients often make repeat visits to the hospital.

CDC Metric:

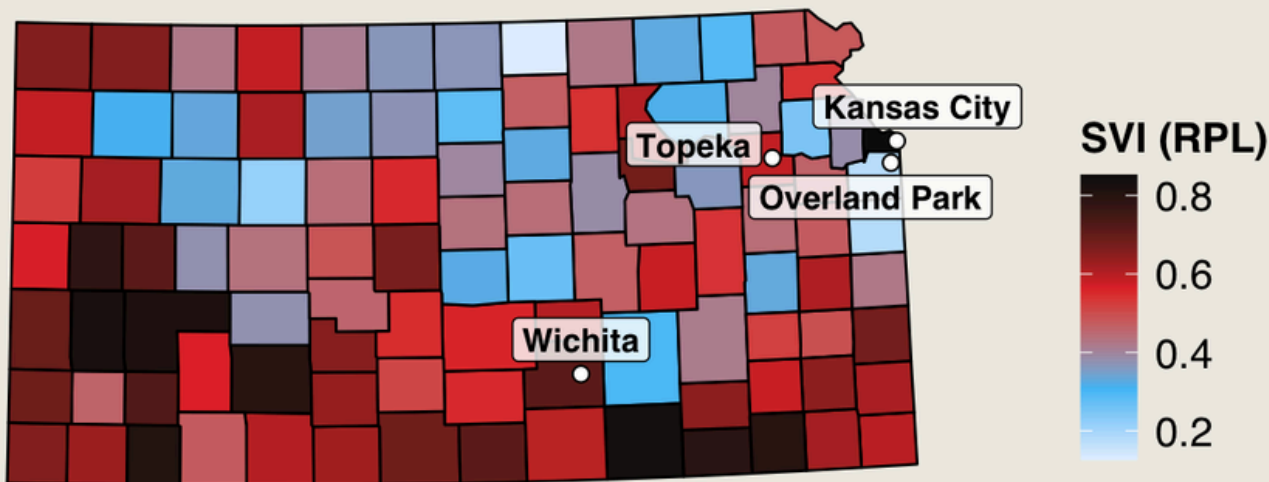
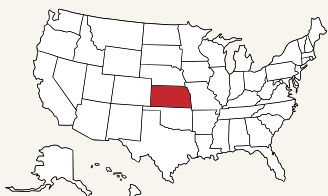
National Center for Health Statistics Data Use



Vulnerable populations

The Geography

Median SVI in Kansas by County

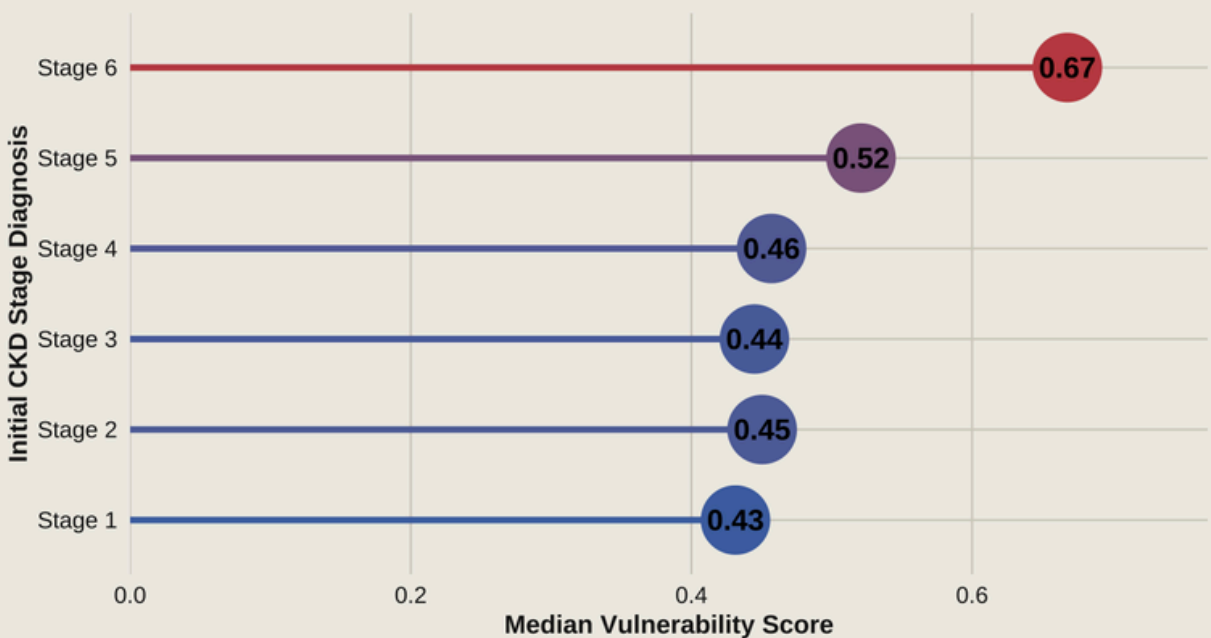


What it suggest about Kansas:

Higher SVI populations are concentrated in **Southern Districts** while **Lowest SVI's** are in **Northern Districts.**

The Diagnosis

Median vulnerability score (SVI) vs initial CKD stage



What it suggest about patients:

The more **socioeconomically vulnerable** the patient, the **sicker** the at first diagnosis.

Patient Journey

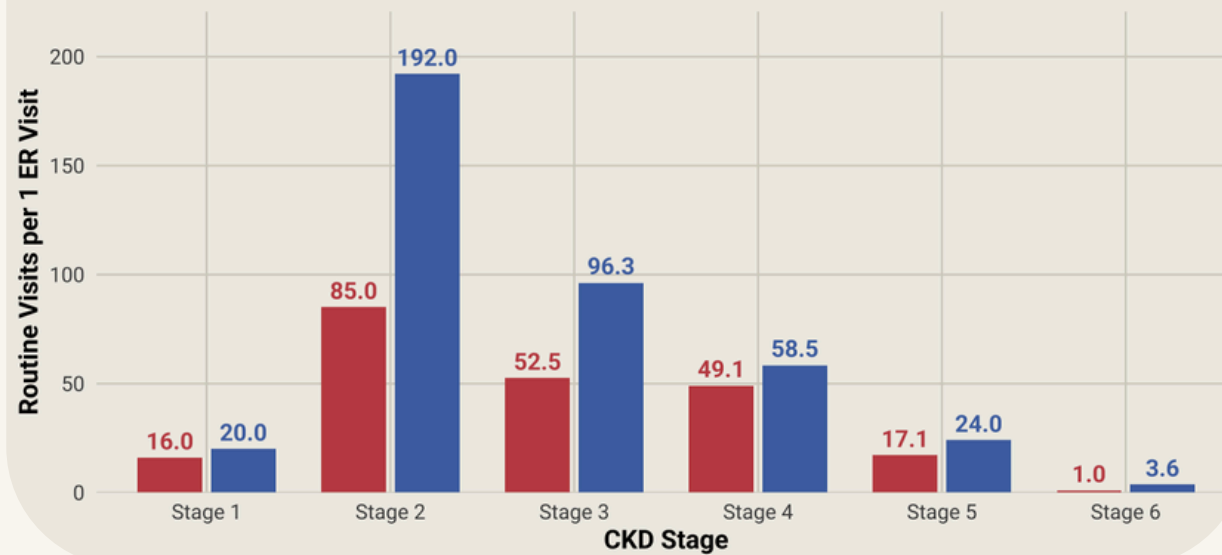
Preventative care inequity



Preventative Care Frequency Across CKD Stages by Vulnerability

Number of Routine Visits (Lab/Office) per 1 Hospital Encounter

CDC Vulnerability Level ■ High ■ Low



What it suggest about patients:

Low vulnerability patients receive **more routine preventive care per ER visit**. High vulnerability patients receive **56% fewer routine visits** per ER visit at stage 2

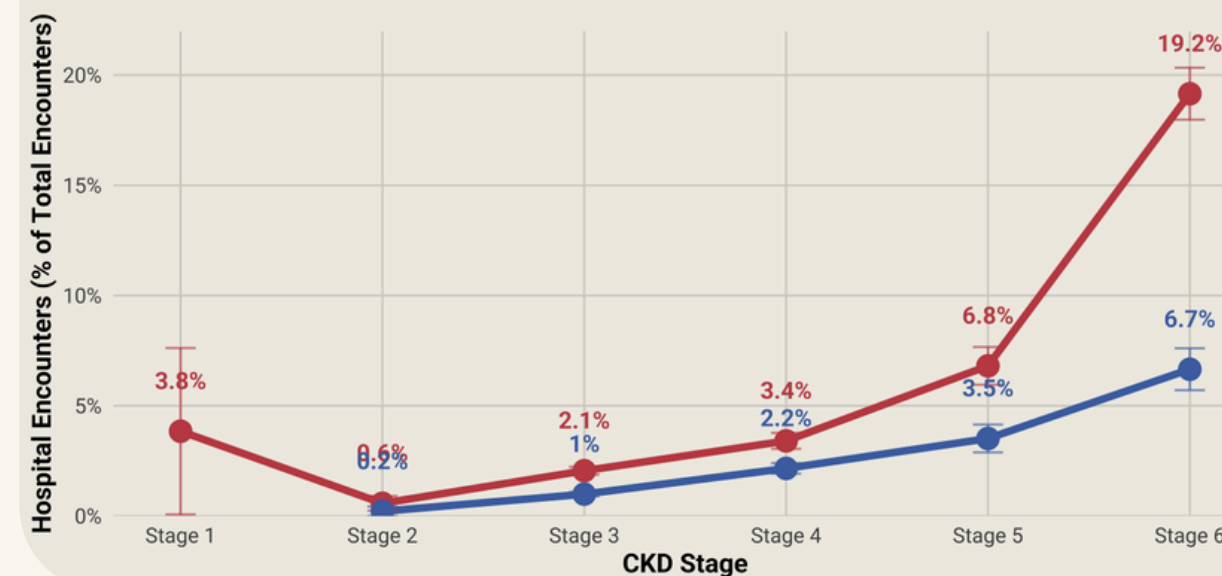
ER Visits by stage



Hospital Encounter Rate Across CKD Stages by Vulnerability

More vulnerable patients have more ER visits at advanced stages

CDC Vulnerability Level ● High ● Low

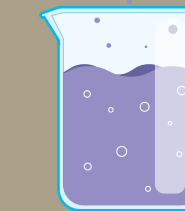


What it suggest about patients:

High vulnerability patients **go to the ER nearly 3x more** than low vulnerability patients **at end-stage CKD (6)** (19.2% vs 6.7%)



LOW SVI patients diagnosed earlier than HIGH SVI



More infrequent preventative care per ER (non elective) encounter



High ER rates for Stage 6 high CDC vulnerability patients

Recommendations:



Expand **telehealth** in high-SVI counties so vulnerable patients can actually **access the routine lab/office** visits they need.



Deploy **mobile kidney health screening units** to high-SVI counties in Southwest Kansas, to **remove transportation and access barriers**.